

HCI and Design

Assignments

Assignment 2 is due TODAY, 11:59pm!

Submit one per pair on Blackboard.

Today

Paper prototyping

- An essential tool in your design toolbox!



How do we design things that actually fit user needs?

Problem:

- We can't evaluate a design until it's built
But...
- After building, changes to the design are difficult
- What to do?

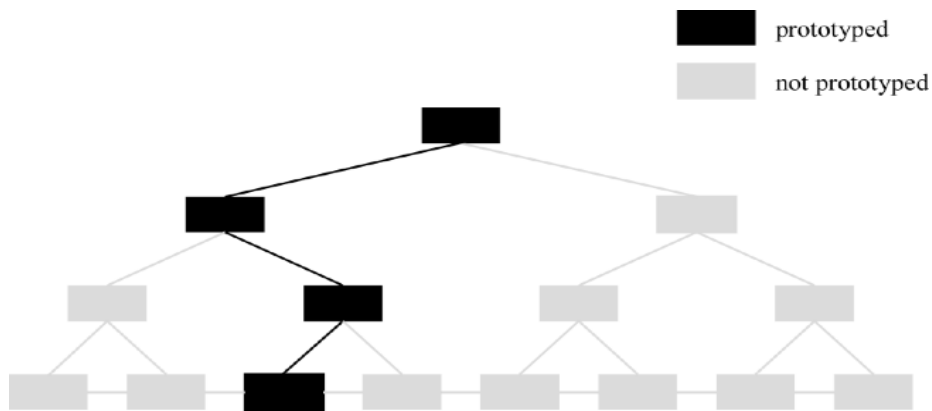
Solution

- Prototype!

Prototyping

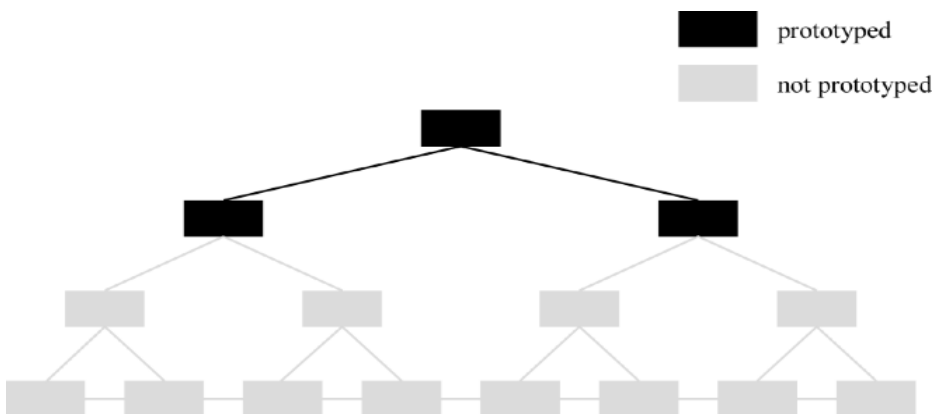
- Simulate the design in low-cost manner
- Make it fast. Make it cheap.
- Facilitate iterative design and evaluation
 - Your first idea is rarely your best!
- Promote feedback
- Allow lots of flexibility for radically different designs
 - Don't kill crazy ideas!

How to prototype?



Vertical - “Deep” prototyping

- Show only portion of interface, but large amount of those portions



Horizontal - “Broad” prototyping

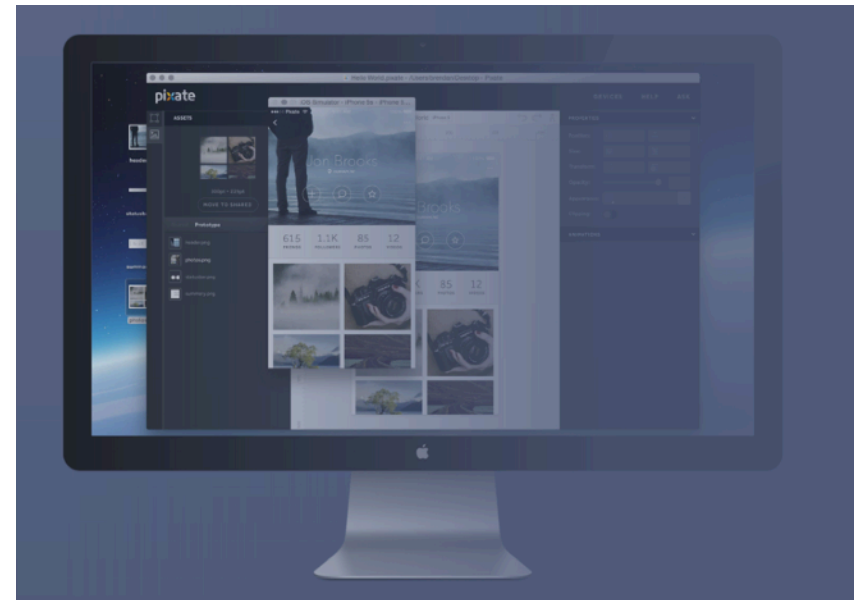
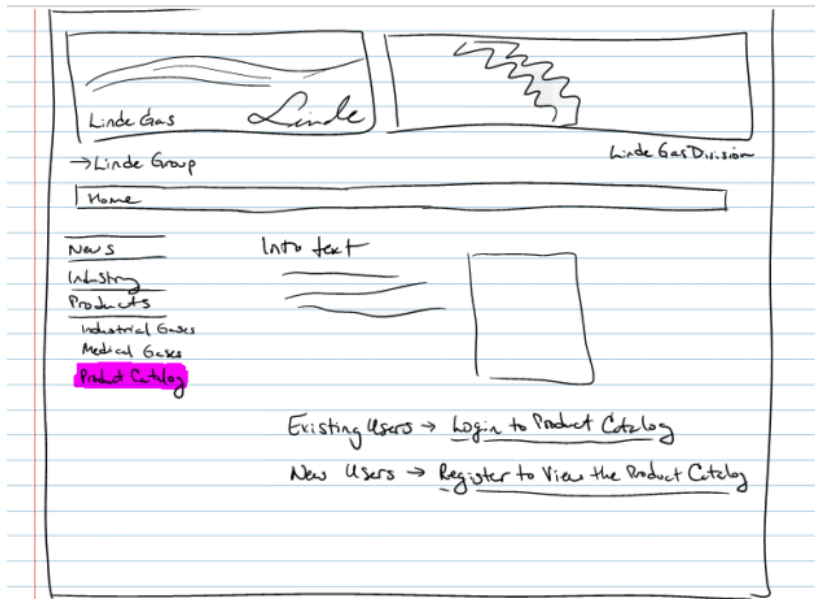
- Show much of the interface, but in a shallow manner

How to prototype?

Low fidelity

vs.

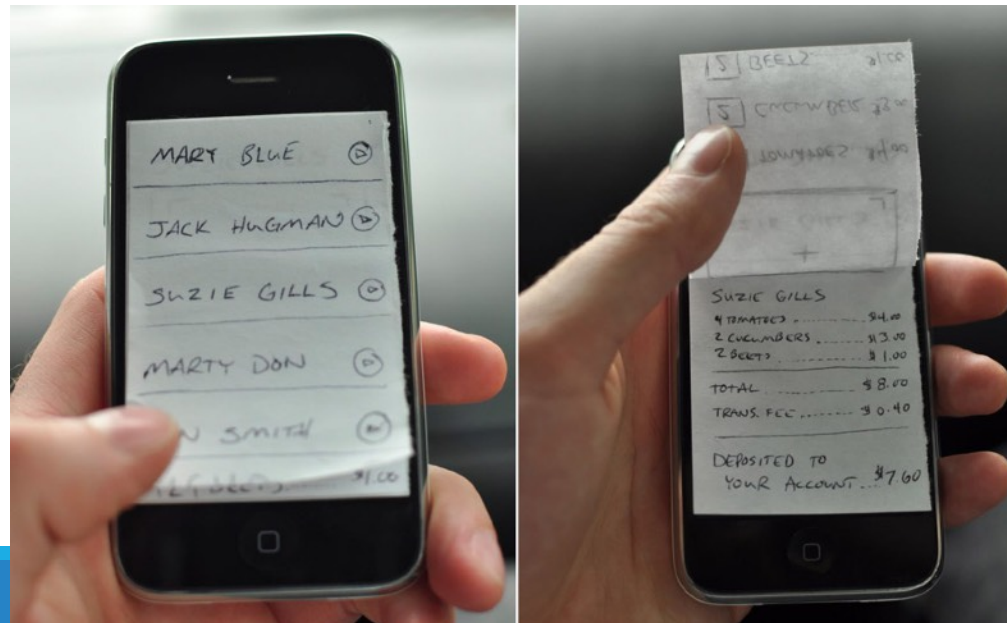
High fidelity



Amount of polish should reflect maturity of the prototype... Why?

“Mixed” fidelity

- Easy access to cameras makes it easy to blur the lines between lo-fi and hi-fi prototypes
- Photos of hand-drawn prototypes can easily be captured and displayed on real screens
- Sequences of photos can also be animated to simulate interaction



Today: Paper-Prototyping

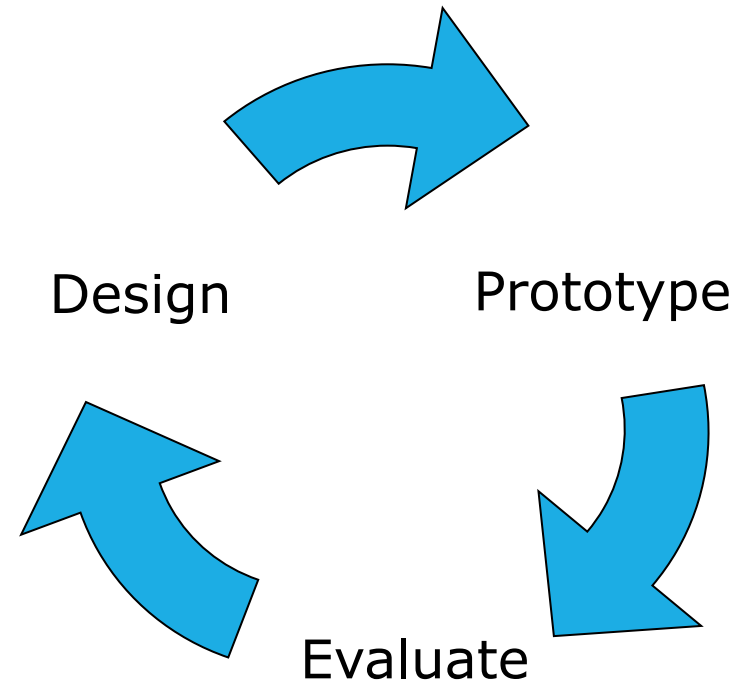
An iterative design method where potential users perform realistic tasks by interacting with a paper version of the interface that is manipulated by a person 'playing computer,' who doesn't explain how the interface is intended to work.



Why do it?

Principle of iterative design

- Quality is partially a function of the number of iterations and refinements that a design undergoes



Why do it?

Low cost

Fast to implement

- Typical hi-fi prototype takes a few weeks as opposed to a paper prototype that takes a few hours

Allows you to merge the design and prototyping phase together

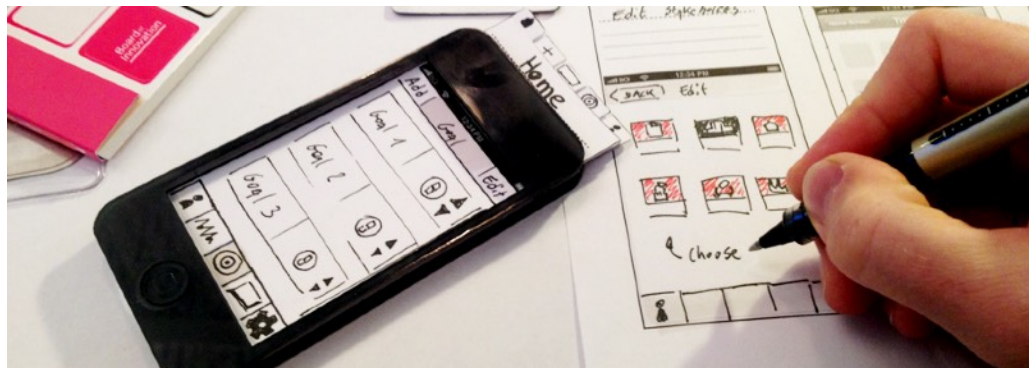
It gets everyone involved!

- Builds teamwork in groups with diverse skill sets
- So simple, no one gets left out

Why do it?

Feedback on the BIG things

- Lo-fi nature forces users to consider usability issues related to layout and control
- Nit picking over choice of colors, buttons sizes, font choice ignored
- Focus on *Content* as opposed to *Appearance*



Why not to do it?

May seem unprofessional to some users

- *Maybe not the right prototype for the VCs* 😊

Can't represent some effects with paper

Typically, you would start with several rounds of paper prototyping, and move towards high-fidelity prototyping as the design becomes more finalized. i.e., YOU DO BOTH!

Building a lo-fi prototype

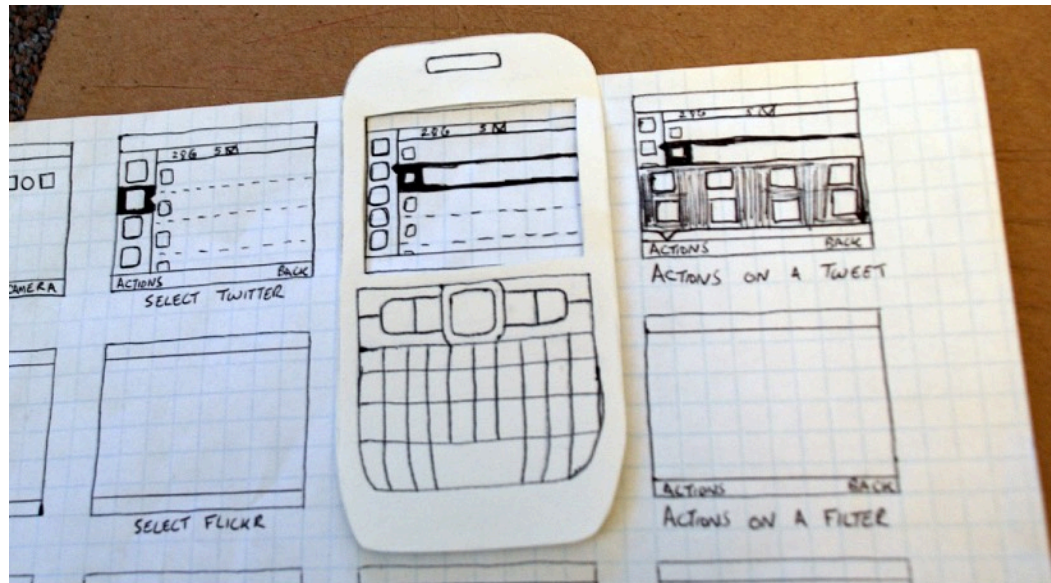
Gather essential materials

- White paper
- 5 by 8 inch cards
- Adhesives
- Markers
- Sticky pads
- Scissors
- Anything else you think of!!!

Building a lo-fi prototype

Don't get carried away with design!

- Goal is to get as much user feedback as possible
- Set a deadline - forget minor details

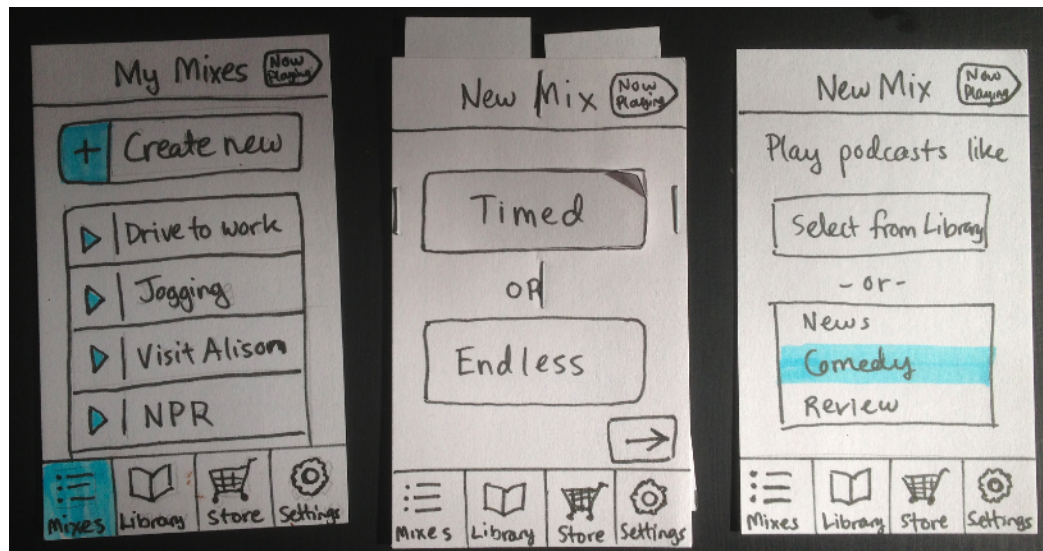


Building a lo-fi prototype

Draw generic frames

Make everything needed to simulate effects

Photocopier/camera is your friend!



Preparing for a test

Select users

- Perform user and task analysis
- Find out educational background, knowledge of computers, typical tasks required
- Get testers who fit the final user profile



Preparing for a test

Prepare test scenarios

Practice

- Sort out bugs/hitches before the real testing
- Get everyone comfortable with their role



Conducting a test

Facilitator

- Encourage user to express thoughts
 - (don't influence decisions!!!!)
- Giving instructions
- Making sure timing is met



Conducting a test

“Computer” person

- Arranges the paper prototype according to user input
- Needs to be organized
- Knows the prototype well
- Make changes quickly



Conducting a test

Observers

- Take notes
- Write possible solutions to problems faced
- Cannot react to user's actions



Evaluating results

Summarize problems (e.g., make a list)

- Usability issues
- Missing (or mis-specified) functional requirements
- Preferences for different alternatives
- User priorities
- Issues outside the user interfaces (e.g., high-level understanding)

Prioritize problems

Construct revised prototype

Iterate, iterate, iterate!

Summary: Paper Prototyping

- An important prototyping tool (but not the only tool!)
- Quick to build/refine, thus enabling rapid design interactions. Useful tool for speeding up the process of iterative design
- Requires minimal resources and materials (cheap!)
- Detects usability problems at a very early stage before implementation.
- Focus on the “right” things early on
- Promotes communication between stakeholders. Team members gain understanding of user needs and priorities
- I recommend you always do a few rounds of paper prototyping for every new design/app/system/solution that you create!